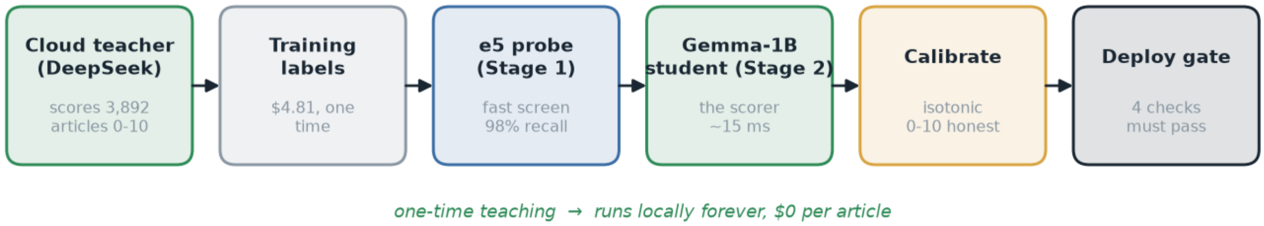
nature_recovery v4

A local, distilled scoring model — and why it beats buying one

Deploy v4 — it beats the incumbent on every metric that matters.



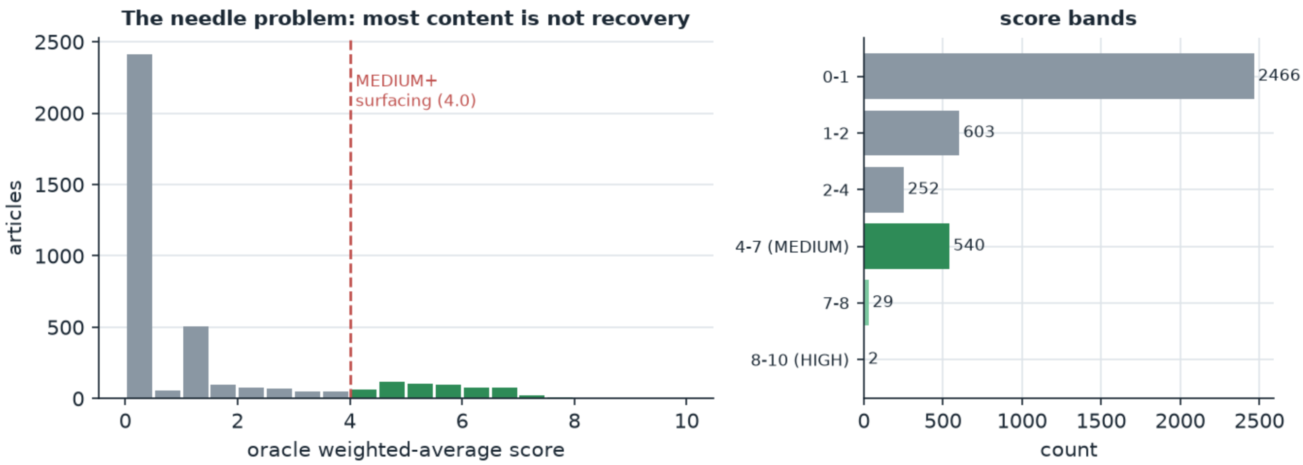
This report walks the whole pipeline from left to right. Every section starts in plain words, then gives the engineering detail underneath.

1. The big picture

In plain words: Cloud AI models are smart but expensive — you pay every single time you ask them a question. We use the expensive model **once** as a teacher: it grades a few thousand news articles for how strongly they show *nature recovering*. Then we train a tiny model on our own computer to imitate the teacher. The tiny model — the “student” — then does the job forever, on our hardware, for essentially $0 per article.

This is **knowledge distillation**. The teacher (DeepSeek, a cloud LLM) produces 0–10 scores on six dimensions of ecological recovery. A Gemma-3-1B student with a LoRA adapter learns to reproduce those scores. At inference we add a fast **e5 embedding probe** that screens out obvious non-matches before the student runs, plus **isotonic calibration** so the numbers stay honest. Finally an automated **agreement gate** must pass before anything ships.

Why this filter is hard: nature-recovery stories are a *needle in a haystack*. In our data only **15%** of articles are genuine recovery; the rest are doom, pledges, or unrelated. A model that lazily says “no” to everything looks accurate but is useless. That single fact drives every design choice below.



2. Why build this instead of buying it

In plain words: You could pay a cloud API or a vendor to score every article. But you'd pay again for every article, forever; you'd send your content to someone else's servers; and you'd get a *general* model that was never tuned to *your* definition of 'recovery'. Our student is cheaper at scale, private, offline-capable, and tuned to exactly the editorial line we want.

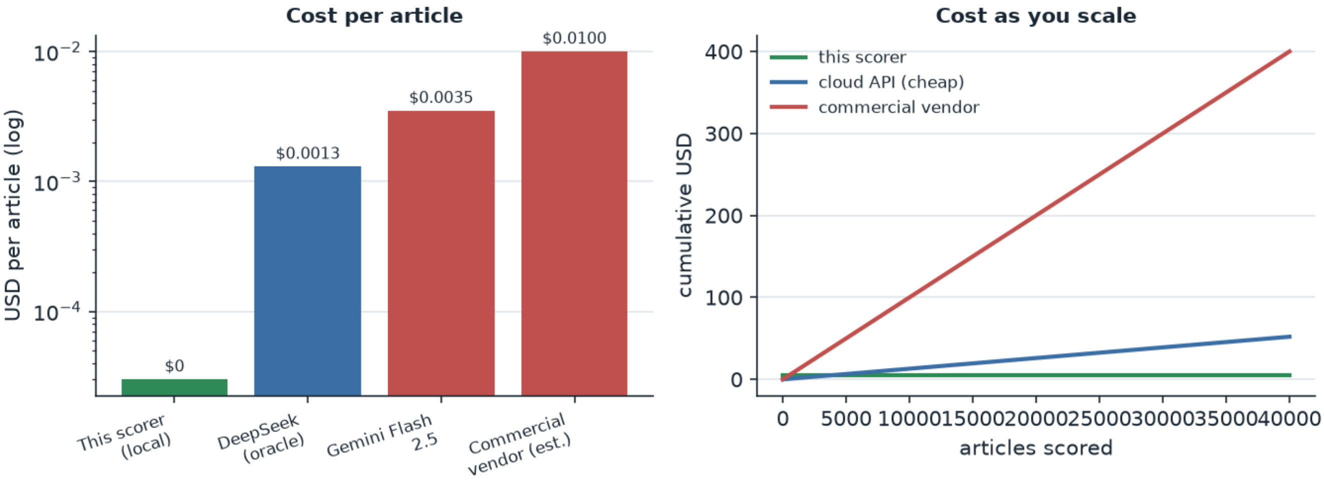
Cost at scale. One-time teaching cost was **$4.81**. After that, scoring is local and free. A cloud API at even $0.0013/article costs more than the entire training run after ~3,700 articles — and we score far more than that every week.

Latency. ~15 ms/article locally vs a network round-trip (100–800 ms) to a cloud API, with rate limits.

Privacy & control. Articles never leave our infrastructure. No third-party retention, no ToS surprises.

Specialisation. Off-the-shelf sentiment/topic APIs don't know that a *pledge* is not a *delivered* protected area. Ours is trained on that exact distinction (#70).

No lock-in / drift control. We own the weights, the calibration, and the gate. We decide when the definition changes — not a vendor pushing a silent model update.



3. Stage 1 — the teaching signal (oracle labels)

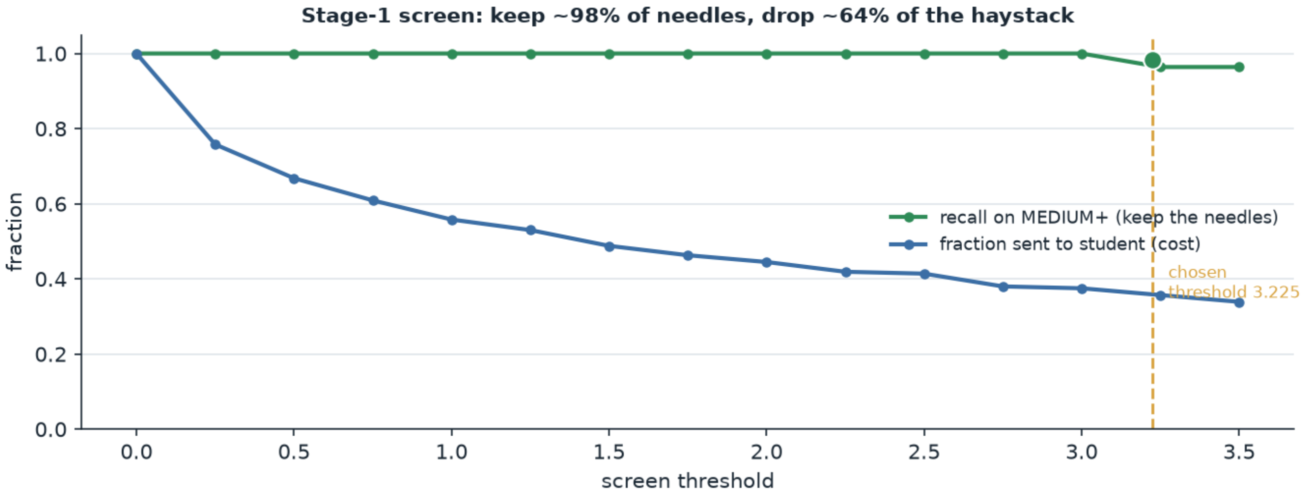
In plain words: First we need an answer key. We asked a strong cloud model to grade 3,892 articles on six things: is nature actually recovering, is there hard data, how ecologically important, how big/long, did humans cause it, and will it last. Those grades become the homework the student learns from.

The oracle outputs **scores only** (0–10 per dimension), never tier labels — so we can change surfacing thresholds later without re-labelling. A **gatekeeper** rule keeps articles with no real recovery evidence from surfacing on the strength of the other dimensions alone. v4's key change (#70): *delivered* structural protection (an enacted marine protected area, an in-force fishing ban, a removed dam) now counts as recovery-in-progress, while mere pledges stay low.

4. Stage 2 — the fast screen (e5 probe)

In plain words: Running the student on every article costs time. So first a much smaller, faster model takes a quick look and throws out the obvious ‘definitely not recovery’ articles. The trick: it must almost never throw out a real one. We tuned it to keep ~98 of every 100 real recovery stories while still skipping ~64% of the junk.

The probe is an e5-small multilingual embedding + a small MLP. Crucially it is trained **recall-first**, not as a regression: the screen threshold is chosen from the validation recall curve at a target false-negative budget, not by minimising average error (which would collapse to a floor predictor on this data). Result: validation recall **98.2%** on MEDIUM+ at threshold 3.225, routing only **36%** of articles to the student.

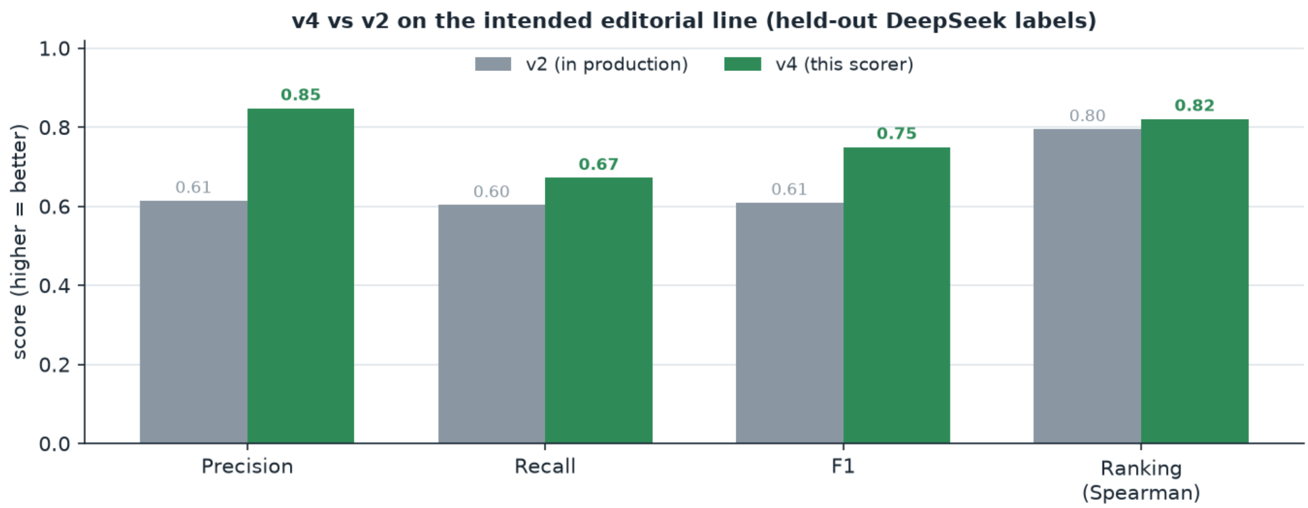


5. Stage 3 — the student model

In plain words: This is the main scorer: a 1-billion-parameter model with a small trained adapter. We judge it not by ‘average error’ (misleading here) but by whether it ranks the right stories to the top — the ones an editor would actually want to see.

Trained with score-based sample weighting (scale 2.0) to counter the class imbalance, and judged on ranking, not average error. The real test is head-to-head against the version in production, both scored against the same held-out ‘answer key’ from the teacher (the editorial line we chose).

On that footing, **v4 beats v2 on every metric**: precision **0.85** vs 0.61, recall **0.67** vs 0.60, F1 **0.75** vs 0.61, ranking (Spearman) 0.82 vs 0.80. The precision gap is the headline: the old v2 (trained on a more *generous* teacher) over-surfaces — a third of what it shows isn't really recovery. v4 is far more precise and still slightly higher recall.

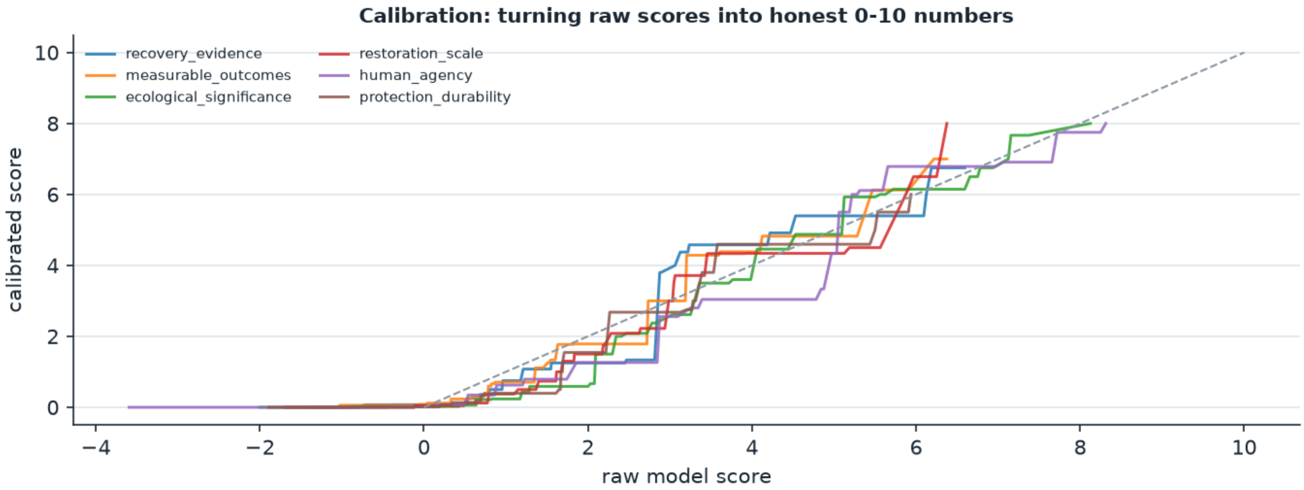


Note on the top band: only 2 training articles score 8–10, so the very top of the scale is barely trainable — we clip rather than over-fit it (documented limitation).

6. Stage 4 — calibration

In plain words: A model's raw output isn't always on a human-meaningful scale — a raw '5' might really mean '6.5'. Calibration is a simple correction curve, fitted on held-out data, that lines the numbers back up with reality so a '7' means what a person would call a 7.

We fit per-dimension **isotonic regression** on the validation set (monotonic, non-parametric). Because only ~2 articles sit in the 8–10 band, we clip the top rather than fitting a spline through 2 points — avoiding wild extrapolation.

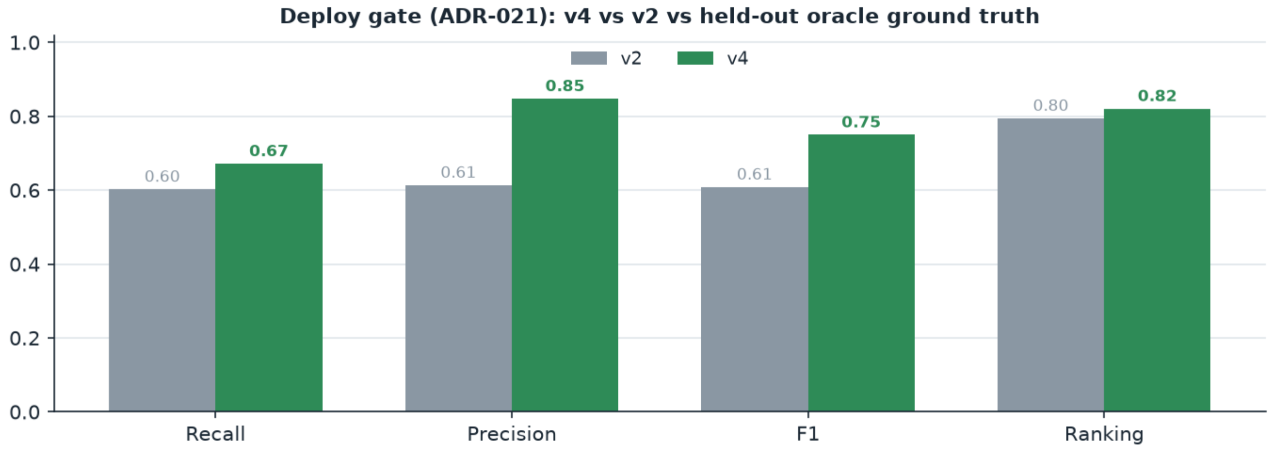


7. Stage 5 — the deploy gate (and how it nearly lied)

In plain words: Before we let the new scorer replace the old one, it must prove — automatically — that it's actually better. But this is also a story about *how* we check, because our first gate gave the wrong answer, and catching that is exactly why you can trust the final result.

The gate that cried wolf. Our first automatic gate compared v4 against the *old model* (v2) as the yardstick, and reported **FAIL**. Before believing it, we opened the actual articles — and found the gate's reference answers were secretly graded by a *different, more generous* teacher (v2's old one). v4 was being punished for correctly rejecting content that teacher over-rated (a corporate 'changemaker' profile, a how-to listicle). 9 of 12 'errors' were v4 being right. The gate wasn't measuring quality — it was measuring disagreement with a bias we deliberately moved away from.

The fix (ADR-021). Judge each model against a held-out 'answer key' from the teacher we actually chose — not against the previous model. On that honest footing:

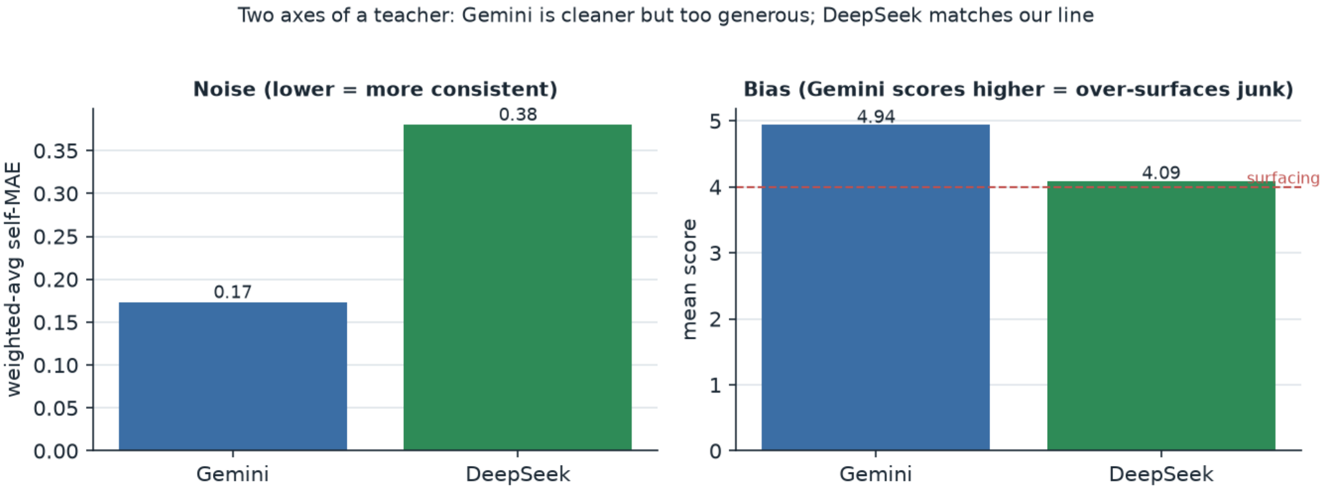


PASS — v4 beats v2 on every metric. The reproduce-don't-assess habit that caught the false alarm is the real reason to trust this number.

8. The teacher matters more than the model

In plain words: The student can only be as good as the teacher it learns from. And a teacher can fail two ways: it can be *inconsistent* (grades the same essay 6 one day, 8 the next), or it can be *biased* (consistently too generous). These are different problems, and confusing them is expensive.

We compared two candidate teachers. One (Gemini) was **2.2× more consistent** — and it was tempting to switch to it. But it was also systematically *generous*: it scored a corporate ‘sustainability changemaker’ profile a 5.6, and a how-to listicle a 5.6 — exactly the content this filter exists to reject. Switching to the ‘cleaner’ teacher would have re-graded the whole training set toward the wrong editorial line. We kept the conservative teacher (DeepSeek) and accept its mild inconsistency, because **bias is the axis that matters**.



The lesson, in one line: a clean, consistent, *wrong* teacher looks like progress and isn’t. Choose the teacher for its judgement first; reduce inconsistency by averaging, never by switching to one that judges differently.

9. Recommendation

Use the local distilled scorer, v4. Judged against the intended editorial line (held-out DeepSeek labels), v4 beats the deployed v2 on recall, precision, F1, ranking, and error — with far higher precision (0.85 vs 0.61: v2 over-surfaces). It also fixes the multilingual recall bug (21.6%→0%) and adds delivered-protection scope (#70). One-time teaching cost was $4.81; scoring is local and free forever.

Question	This scorer	Buying it
Metric (vs DeepSeek line)	v4 (this scorer)	v2 (incumbent)
Precision	0.848	0.614
Recall	0.672	0.603
F1	0.750	0.609
Ranking (Spearman)	0.821	0.795
Cost/article	$0 local	$0 local
Multilingual recall bug	fixed (0%)	21.6% dropped

The teaching cost is already spent. Every article from here is free, private, fast, and tuned to exactly the conservative editorial line we chose — and this version is measurably better than what's in production. That is the case for using this scorer, and for shipping v4.